TurboQuant v1.5.2+

Work Log — Session 3 (PM)

2026-04-08 Afternoon

tbq4 D=512 Fix + head_dim=64 Cross-Head WHT Breakthrough

4-bit WHT Domain Mismatch Fix

MMSE Dynamic Softening

QJL Noise Discovery

Pauli Test Passed

AmesianX

Session Overview

- `PM` — tbqp4/tbq4 testing — garbage output found, ran math bench
- `PM` — Root cause: 512-point WHT missing in tbq4_0/tbqp4_0 D=512 encoding
- `PM` — Fix: added quantize_f32_tbq4_0_block_512 + tbqp4_0_block_512
- `PM` — tbqp4/tbq4 math bench 20/35 = tied with f16!
- `PM` — PPL bench: tbqp4/tbq4 1.08x vs f16 (tbqp3/tbq3 1.19x on same file)
- `Eve` — head_dim=64 challenge — cross-head WHT retry
- `Eve` — global L2 norm → per-head RMS norm fix
- `Eve` — Dynamic sharpening _2/_3 attempt → degraded (SQNR values inappropriate)
- `Night` — MMSE theory: D=64 needs **softening**, not sharpening!
- `Night` — K=f16 V=tbq3 experiment → perfect (V is not the problem, K is the bottleneck)
- `Night` — Key finding: QJL is the primary source of K score noise at D=64!
- `Night` — tbqp3 → tbq3 auto-switch (QJL removed) → **Pauli test passed!**

tbq4/tbqp4 D=512 Fix

Bug Fix

PROBLEM

- tbq4_0/tbqp4_0 encoding: only performed 256-point WHT
- Q preprocessing: performed 512-point WHT
- WHT domain mismatch → garbage output
- tbq3 already had the 512 version

FIX

- Added `quantize_f32_tbq4_0_block_512`
- Added `quantize_f32_tbqp4_0_block_512`
- set-rows.cu: head_dim≥512 dispatch
- Math bench 20/35 = tied with f16 ✓

Cross-Head WHT Breakthrough

Breakthrough

1. PER-HEAD RMS NORM

- Before: global L2 norm (shared across 8 heads)
- Problem: high-energy heads dominated the norm
- Low-energy heads lost information → garbage
- Fix: independent RMS norm per head

2. QJL NOISE DISCOVERY

- TBQP3: 2-bit MSE + 1-bit QJL
- QJL's random projection → K score noise
- D=256: noise is small (negligible)
- **D=64: noise exceeds signal gap!**

Solution: Auto-switch K from tbqp3 → tbq3 at head_dim=64. Invest all 3 bits in Lloyd-Max centroids. SQNR improved 2.3x (3.45 → 7.8).

MMSE Dynamic Softening

Theory

D≥256: SHARPENING (A > 1)

- High SQNR → correct token almost always ranks first
- softmax slightly flattened → corrected by α
- `α(N) = 1 + c × √(ln N)`

D=64: SOFTENING (A < 1)

- Low SQNR → noise flips rankings
- Sharpening goes all-in on the wrong answer
- `α(N) = SQNR / (SQNR + √(ln N / ln N₀))`

TYPE	SQNR	A(N=10)	A(N=2048)	A(N=131K)
TBQP3 _2/_3	3.45	0.863	**0.775**	0.735
TBQ3 _2/_3	7.80	0.934	**0.886**	0.863
TBQ4 _2/_3	14.05	0.963	**0.934**	0.919

K vs V Isolation Test

Verification

K	V	PAULI TEST	CONCLUSION
f16	f16	✓ Perfect	Baseline
f16	tbq3_2	✓ Perfect	**V is not the problem**
tbqp3_3	f16	✗ Loop	**K is the bottleneck**
tbqp3_3	tbq3_2	✗ Loop	K noise confirmed
tbq3_3	tbq3_2	✓ Passed!	**QJL removal = solved**

Conclusion: V is fine at 3-bit. Removing QJL from K and using pure 3-bit Lloyd-Max makes head_dim=64 work.

TBQP3 vs TBQ3 K Score Quality

Math

TYPE	STRUCTURE	SCORE SQNR (D=64)	NOISE/SIGNAL
TBQ3	3-bit Lloyd-Max (8 centroid)	**7.8**	**35.8%**
TBQP3	2-bit MSE + 1-bit QJL	**3.45**	**53.8%**

Key insight: QJL's 1-bit random projection reduces K reconstruction MSE but adds a **second noise source** to attention scores. At D=256 this is negligible, but at D=64 it exceeds the signal gap (~1.0) with noise of 1.67. With TBQ3, investing all 3 bits in centroids keeps max noise at 1.11, within the gap.

Final Results Achieved

- tbq4/tbqp4 D=512: WHT domain mismatch fixed, math bench 20/35 = tied with f16
- head_dim=64 cross-head: quantization quality restored with per-head RMS norm
- K QJL noise discovered: tbqp3 → tbq3 auto-switch improved SQNR 2.3x
- MMSE dynamic softening: unified theory for D≥256 sharpening / D=64 softening
- GPT OSS 120B (head_dim=64) Pauli test **passed**
- Release workflow fixed (master→main, per-architecture builds)

D=64
Cross-head Working

2.3x
SQNR Improvement (QJL Removed)

α<1
MMSE Softening

PASS
Pauli Test